

Yuankun Xie

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RESEARCH INTERESTS

Audio Deepfake Detection; Domain Generalization; Out-of-Distribution (OOD) Detection; Audio Large Language Models (ALLMs); Neural Audio Codec

EDUCATION

- **Institute of Automation, Chinese Academy of Sciences (CASIA)** 2023.9 – 2026.7
Joint Ph.D. Program in Information and Communication Engineering, Advisor: Prof. Jianhua Tao Beijing, China
- **Communication University of China** 2022.9 – 2026.7
Ph.D. Candidate in Information and Communication Engineering, Advisor: Prof. Long Ye Beijing, China
- **Communication University of China** 2018.9 – 2021.7
M.Eng. in Information and Communication Engineering, Advisor: Dr. Yutian Wang Beijing, China
- **North China University of Technology** 2013.9 – 2017.7
B.Eng. in Communication Engineering Beijing, China

RESEARCH EXPERIENCE

- **Open-Domain Audio Deepfake Detection** 2022.9 – Present
Ph.D. Research
 - **Research Scope:** Global detection, partial manipulation localization, and deepfake algorithm attribution in open-domain scenarios.
 - **Data-centric Contributions:**
 - * **Codecfake:** First dataset constructed from neural audio codec generation mechanisms for ALM-based deepfake detection, exceeding 10,000 downloads.
 - * **FSD:** First work to formalize *fake singing voice* detection, releasing a Chinese fake song dataset and benchmark.
 - * **FSW:** First large-scale real-world Chinese social-media speech deepfake dataset with platform-level evaluation protocols.
 - * **ST-Codecfake:** First open-set neural codec source tracing dataset with ID/OOD evaluation protocols.
 - **Algorithmic Contributions:**
 - * **ASDG:** Domain-generalized representation learning via real-domain aggregation and fake-domain dispersion.
 - * **WPT:** Wavelet Prompt Tuning for all-type audio (speech, sound, singing, music), improving generalization with parameter-efficient training.
 - * **TDL:** Efficient frame-level localization for partially spoofed audio, achieving SOTA on ASVspoof19-PS.
 - * **REFD:** Two-stage source tracing with score-based OOD detection, achieving SOTA on ADD2023 Track 3.

INDUSTRIAL RESEARCH INTERNSHIPS

- **Ant Group** 2025.10 – 2026.5
Research Intern Beijing, China
 - Proposed Frequency-Time GRPO (FT-GRPO) for interpretable all-type audio deepfake detection with Audio LLMs via reinforcement learning.
 - Co-organized the ACM Multimedia 2026 All-Type Audio Deepfake Detection (AT-ADD) Challenge.
- **ByteDance** 2025.6 – 2025.7
Research Intern Beijing, China
 - Spatial audio synthesis for First-Order Ambisonics (FOA).
 - Reproduced SOTA models for FOA generation from 360-degree videos.
 - Introduced text-based spatial descriptions to enhance spatial realism and audio-video alignment; deployed in production.
- **Tencent YouTu Lab** 2024.11 – 2025.4
Research Intern Beijing, China
 - Built FSW dataset from real social-media platforms for wild deepfake detection.
 - Proposed codec-robust detection methods adaptable to diverse platform compression conditions.
 - Developed open-set neural codec attribution for ALM-generated audio; deployed on Tencent Cloud.
 - Conducted all-type (speech, sound, singing, music) cross-type evaluation and prompt-based SSL training.
- **Tsinghua Qiyuan Lab** 2023.9 – 2024.11
Research Intern Beijing, China
 - Designed domain-invariant representations via real aggregation and fake dispersion for robust detection.
 - Proposed efficient partial spoof localization with frozen SSL encoders.
 - Conducted deepfake source tracing for unknown generation algorithms.

First or co-first author of 14 papers in top-tier venues (TIFS, TASLP, ICASSP, INTERSPEECH, AAI), with 10+ additional collaborative publications.

- [J1, TIFS 2024] **Yuankun Xie**, Haonan Cheng, Yutian Wang, Long Ye. *Domain Generalization via Aggregation and Separation for Audio Deepfake Detection*.
- [J2, TASLP 2025] **Yuankun Xie**, Yi Lu, Ruibo Fu, Zhengqi Wen, Zhiyong Wang, Jianhua Tao, Xin Qi, Xiaopeng Wang, Yukun Liu, Haonan Cheng, Long Ye, Yi Sun. *The Codefake Dataset and Countermeasures for Universal Detection of Deepfake Audio*.
- [C1, AAI 2026] **Yuankun Xie**, Ruibo Fu, Zhiyong Wang, Xiaopeng Wang, Songjun Cao, Long Ma, Haonan Cheng, Long Ye. *Detect All-Type Deepfake Audio: Wavelet Prompt Tuning for Enhanced Auditory Perception*.
- [C2, ICASSP 2026] **Yuankun Xie**, Ruibo Fu, Xiaopeng Wang, Zhiyong Wang, Ya Li, Zhengqi Wen, Haonan Cheng, Long Ye. *Fake Speech Wild: Detecting Deepfake Speech on Social Media Platforms*.
- [C3, ICASSP 2024] **Yuankun Xie**, Haonan Cheng, Yutian Wang, Long Ye. *An Efficient Temporal Deepfake Localization Approach Based on Embeddings for Partially Spoofed Audio Detection*.
- [C4, ICASSP 2024] **Yuankun Xie**, Jingjing Zhou, Xiaolin Lu, Zhenghao Jiang, Yuxin Yang, Haonan Cheng, Long Ye. *FSD: An Initial Chinese Dataset for Fake Song Detection*.
- [C5, INTERSPEECH 2024] **Yuankun Xie**, Ruibo Fu, Zhengqi Wen, Zhiyong Wang, Xiaopeng Wang, Haonan Cheng, Long Ye, Jianhua Tao. *Generalized Source Tracing: Detecting Novel Audio Deepfake Algorithms with Real Emphasis and Fake Dispersion Strategy*.
- [C6, INTERSPEECH 2024] Yi Lu[†], **Yuankun Xie**[†], Ruibo Fu, Zhengqi Wen, Jianhua Tao, Zhiyong Wang, Xin Qi, Xuefei Liu, Yongwei Li, Yukun Liu, Xiaopeng Wang, Shuchen Shi. *Codefake: An Initial Dataset for Detecting LLM-based Deepfake Audio*.
- [C7, INTERSPEECH 2024 (ASVspoof 5)] **Yuankun Xie**, Xiaopeng Wang, Zhiyong Wang, Ruibo Fu, Zhengqi Wen, Haonan Cheng, Long Ye. *Temporal Variability and Multi-Viewed Self-Supervised Representations to Tackle the ASVspoof 5 Deepfake Challenge*.
- [C8, INTERSPEECH 2023] **Yuankun Xie**, Haonan Cheng, Yutian Wang, Long Ye. *Learning a Self-Supervised Domain-Invariant Feature Representation for Generalized Audio Deepfake Detection*.
- [C9, IJCAI 2023 Workshop] **Yuankun Xie**, Haonan Cheng, Yutian Wang, Long Ye. *Single Domain Generalization for Audio Deepfake Detection*.
- [C10, ICME 2023] Yutian Wang[†], **Yuankun Xie**[†], Kun Zhao, Hui Wang, Qin Zhang. *Unsupervised Quantized Prosody Representation for Controllable Speech Synthesis*.
- [C11, ISCSLP 2024] **Yuankun Xie**, Chenxu Xiong, Xiaopeng Wang, Zhiyong Wang, Yi Lu, Xin Qi, Ruibo Fu, Yukun Liu, Zhengqi Wen, Jianhua Tao, et al. *Does Current Deepfake Audio Detection Models Effectively Detect ALM-based Deepfake Audio?*.
- [S1, ACM MM 2026] **Yuankun Xie**, Xiaoxuan Guo, Jiayi Zhou, Tao Wang, Jian Liu, Ruibo Fu, Xiaopeng Wang, Haonan Cheng, Long Ye, et al. *Interpretable All-Type Audio Deepfake Detection with Audio LLMs via Frequency-Time Reinforcement Learning*.
- [S2, ACM MM 2026] Xiaoxuan Guo[†], **Yuankun Xie**[†], Haonan Cheng, Jiayi Zhou, Jian Liu, Hengyan Huang, Long Ye, Qin Zhang, et al. *Towards Explicit Acoustic Evidence Perception in Audio LLMs for Speech Deepfake Detection*.
- [S3, Neurocomputing] **Yuankun Xie**, Xiaopeng Wang, Zhiyong Wang, Ruibo Fu, Zhengqi Wen, Songjun Cao, Long Ma, Chenxing Li, Haonan Cheng, Long Ye. *Neural Codec Source Tracing: Toward Comprehensive Attribution in Open-Set Conditions*.

COMPETITIONS

- [1] ICASSP 2026 Environmental Sound Deepfake Detection (ESDD) Track 1 & Track 2: 1st Place
- [2] ICME 2026 Environmental Sound Deepfake Detection 2 (ESDD2): 2nd Place
- [3] Alibaba Tianchi 2025 Global AI Security Challenge (Audio Deepfake): Preliminary 1/360, Final 3/360
- [4] INTERSPEECH 2024 ASVspoof 5 (Progress): 2/48
- [5] IJCAI 2024 FinVolution Cup: Preliminary 2/202, Final 8/30
- [6] IJCAI 2023 Audio Deepfake Detection (ADD) Challenge Track 1.2: 5/52
- [7] IJCAI 2023 Audio Deepfake Detection (ADD) Challenge Track 2: 6/17